

# Albania's Brain Drain: A Low-Level Data-Driven Research Paper Draft

Migration, Employment, Demographic Change and Economic Development

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**Important statement.** *This is a low-level academic research draft. It is not intended for publication. Some findings are preliminary and should be updated after the full ETL, statistical analysis, and forecasting stages are completed.*

**Abstract.** This paper studies Albania's brain drain as a demographic and economic issue connected to migration, labor-market conditions, population decline, and long-term development risk. The study uses public datasets from the World Bank, Eurostat, INSTAT, and the 2024 Albania Household Migration Survey to describe how Albania's population has changed since 2000, where Albanian citizens are moving in Europe, and how residence permits for employment, education, and family reasons can be used as indicators of migration channels. The preliminary descriptive evidence shows a strong population decline, consistently negative net migration, increasing demographic pressure, and a migration pattern that is closely connected to employment opportunities and household economic decisions. The paper does not claim final causal results. Instead, it provides a structured academic framework that can later be expanded into a complete data analytics project with ETL, PostgreSQL, exploratory analysis, correlation testing, forecasting, and Power BI reporting.

**Keywords:** Albania, brain drain, migration, population decline, unemployment, residence permits, demographic change, data analytics.

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## 1 Introduction

Brain drain is the movement of educated, skilled, or working-age people from one country to another in search of better opportunities. For Albania, migration is not only a personal or family decision; it is also a national development issue. When young people, professionals, students, and workers leave the country, Albania can lose labor supply, tax potential, human capital, and future demographic stability. At the same time, migration can also produce positive effects through remittances, international experience, return migration, and diaspora networks.

This paper treats brain drain as a complex issue that combines demographic, economic, and social factors. The main research question is: **What economic and demographic factors are connected to migration and brain drain from Albania?** The study is designed as a low-level academic project, meaning that it contains the basic elements of a research paper but does not claim the depth of a peer-reviewed publication.

The project is also designed as a data analytics portfolio project. Therefore, the study does not only describe the problem in words. It proposes a complete workflow: raw data collection, data cleaning, database design, statistical analysis, forecasting, visualization, and interpretation in business language.

## 2 Background and Context

Albania has experienced major demographic changes over the last three decades. Official census evidence reports that the resident population on 18 September 2023 was 2,402,113, a decline of about 420,000 persons compared with the 2011 Census. The same census summary states that the decline follows a trend observed since 1990 and is linked to emigration. INSTAT's population release for 1 January 2025 reported a population of 2,363,314 inhabitants, a decline of 1.2 percent compared with 1 January 2024. It also reported negative net migration of -28,836 people for 2024, despite a positive natural increase of 1,208.

Migration is especially important because it can change both the size and the structure of the population. If people leaving are mainly young, educated, or working-age individuals, the impact is stronger than a simple decline in total population. The country can experience a smaller labor force, fewer students and professionals, higher dependency pressure, and possible shortages in sectors such as health, technology, construction, services, and education.

The 2024 Albania Household Migration Survey gives important qualitative and quantitative context. It describes Albanian migration as strongly related to economic opportunity, with current migrants concentrated in the 25-44 age group. It also reports that migrants often expect higher income abroad, although their expectations can be larger than their real earnings and savings. This shows that migration decisions are affected by wages, employment conditions, information, networks, and expectations.

## 3 Problem Statement

The problem studied in this paper is that Albania is facing population decline and sustained migration pressure. The concern is not only that people leave the country, but that the people leaving may include young workers, students, and skilled individuals who are important for future economic development. This can create a cycle where weak labor-market opportunities encourage migration, and migration then reduces the domestic human-capital base needed for stronger growth.

The study assumes that brain drain should not be analyzed only as a population problem. It should also be connected to unemployment, GDP growth, GDP per capita, remittances, residence permits, age structure, and destination-country demand. For this reason, a data-driven approach is useful because it can combine multiple indicators and avoid explaining migration from only one perspective.

## 4 Research Aim, Objectives and Questions

### 4.1 Research Aim

The aim of this study is to analyze Albania's brain drain by connecting migration indicators with demographic and economic indicators from 2000 onward, and by building a framework for future forecasting to 2035.

### 4.2 Objectives

- Measure Albania's population decline since 2000.
- Describe net migration trends and identify years with stronger migration pressure.
- Identify important European destination countries for Albanian citizens using Eurostat migration-flow data.
- Analyze residence permits by reason, especially employment, education, family, and other reasons.
- Compare migration patterns with unemployment, GDP growth, GDP per capita, remittances, and FDI.
- Describe changes in youth, working-age, and elderly population groups.
- Develop a basic forecast framework for population and demographic risk to 2035.
- Translate findings into simple policy recommendations.

### 4.3 Research Questions

**RQ1.** Has Albania experienced significant population decline since 2000?

**RQ2.** Which years show stronger migration pressure?

**RQ3.** Where are Albanian citizens migrating in Europe?

**RQ4.** Is migration connected to unemployment and GDP growth?

**RQ5.** How do residence permits explain employment, education, and family migration channels?

**RQ6.** How is Albania's age structure changing?

**RQ7.** What are the likely effects of brain drain on Albania's development?

## 5 Data Sources

The study uses secondary public data. The data package contains international datasets for quantitative analysis and official reports for context and interpretation. The main datasets are summarized below.

**Table 1.** Primary data sources used in the study.

| Source     | Dataset / Report                           | Main Use in the Study                                                                                  |
|------------|--------------------------------------------|--------------------------------------------------------------------------------------------------------|
| World Bank | World Development Indicators for Albania   | Population, net migration, GDP, GDP growth, GDP per capita, unemployment, remittances, FDI             |
| World Bank | Population Estimates and Projections       | Youth population, working-age population, elderly population, dependency pressure, projections to 2035 |
| Eurostat   | Immigration by citizenship                 | Albanian citizens arriving in European reporting countries; used as a proxy for international mobility |
| Eurostat   | Emigration by citizenship                  | Albanian citizens leaving European reporting countries; used for comparison only                       |
| Eurostat   | First residence permits                    | Employment, education, family, and other permit reasons for Albanian citizens                          |
| INSTAT     | Albania Population and Housing Census 2023 | Official census evidence on population decline and demographic structure                               |

| Source              | Dataset / Report                        | Main Use in the Study                                                              |
|---------------------|-----------------------------------------|------------------------------------------------------------------------------------|
| INSTAT              | Population of Albania, 1 January 2025   | Latest official population validation and migration components                     |
| World Bank / INSTAT | Albania Household Migration Survey 2024 | Context on migration motives, employment, wages, remittances, and return migration |

**Note.** Eurostat immigration into European reporting countries is not the same as direct emigration from Albania. It is used as a proxy for Albanian international mobility and must be interpreted carefully.

## 6 Methodology

The methodology follows a basic data analytics research design. First, the datasets are inspected to understand structure, years, missing values, and category definitions. Second, the data is cleaned and reshaped into long format where each row represents one observation, such as country-year-indicator-value. Third, descriptive analysis is used to identify trends in population, migration, permits, unemployment, and GDP. Fourth, statistical tests are planned to examine relationships between migration and economic indicators. Finally, simple forecasting models are planned to estimate demographic pressure to 2035.

### 6.1 Planned Analytical Steps

- Step 1:** Extract the raw CSV, CSV.GZ, and PDF source files.
- Step 2:** Filter the World Bank data for selected indicators only.
- Step 3:** Filter Eurostat data to Albanian citizens only, because files may also contain TOTAL or other citizenship categories.
- Step 4:** Avoid double counting by not adding TOTAL categories together with subcategories.
- Step 5:** Create cleaned datasets for population, economy, unemployment, migration flows, and residence permits.
- Step 6:** Build a master analysis dataset by year.
- Step 7:** Run descriptive analysis and visualizations.
- Step 8:** Run Pearson and Spearman correlations with p-values.
- Step 9:** Use linear regression, moving average, and ARIMA-style models for basic forecasting.
- Step 10:** Write findings in simple business and policy language.

### 6.2 Planned Hypotheses

**Hypothesis 1:** There is a statistically significant relationship between unemployment and net migration in Albania.

**Hypothesis 2:** There is a statistically significant relationship between GDP growth and migration pressure.

**Hypothesis 3:** Residence permits for employment and education are connected to economic and human-capital migration channels.

## 7 Preliminary Descriptive Findings

The findings below are preliminary descriptive results from the available datasets. They are useful for building the paper, but they should be updated after the full cleaning, statistical testing, and dashboard stages.

### 7.1 Population Decline

World Bank data shows that Albania's population decreased from 3,089,027 in 2000 to 2,377,128 in 2024. This is a decline of 711,899 people, or approximately 23.0 percent. This supports the idea that population decline is one of the central problems in the study.

**Figure 1.** Albania population trend, 2000-2024. Source: World Bank WDI dataset.

**Table 2.** Population and demographic indicators, 2000 to latest.

| Indicator                    | 2000      | Latest / Projection | Interpretation                                     |
|------------------------------|-----------|---------------------|----------------------------------------------------|
| Population total             | 3,089,027 | 2,377,128 in 2024   | Decline of 23.0% since 2000                        |
| Youth population 0-14        | 914,868   | 400,327 in 2024     | Major decline in the younger population base       |
| Working-age population 15-64 | 1,947,712 | 1,574,525 in 2024   | Lower labor-force potential than in 2000           |
| Elderly population 65+       | 226,447   | 402,276 in 2024     | Population ageing is increasing                    |
| Projected total population   | -         | 2,155,920 in 2035   | Projection suggests continued demographic pressure |

## 7.2 Net Migration Pressure

The World Bank net migration indicator is negative for the period analyzed. From 2000 to 2025, cumulative net migration is approximately -856,427. The average annual net migration value is around -32,940. The strongest negative value in this period appears in 2000, with -60,531. Even when the negative value becomes smaller in some years, the overall direction remains a warning sign.

**Figure 2.** Albania net migration trend, 2000-2025. Source: World Bank WDI dataset.

## 7.3 European Destination Patterns

Eurostat immigration data, after filtering to Albanian citizens only and excluding aggregate country groups, shows that the strongest available destination totals for 2015-2024 are concentrated mainly in Italy and Germany. In the filtered dataset, total Albanian-citizen immigration into reporting European countries reached 51,732 in 2023, while the 2024 value was 45,423. These figures do not represent all emigration from Albania, because reporting coverage and missing country data can affect the totals.

**Table 3.** Leading European destinations for Albanian-citizen immigration, 2015–2024.

| Rank | Destination country | Total Albanian-citizen immigration, 2015-2024 |
|------|---------------------|-----------------------------------------------|
| 1    | Italy               | 206,053                                       |
| 2    | Germany             | 141,572                                       |
| 3    | Sweden              | 8,601                                         |
| 4    | Croatia             | 7,584                                         |
| 5    | Belgium             | 7,154                                         |

**Note.** The table uses only rows where citizenship is Albania and excludes EU aggregate categories. Some important destinations may be missing or incomplete in the available Eurostat extract.

## 7.4 Residence Permits by Reason

Eurostat first residence permit data for EU27 in 2024 shows 66,161 first residence permits issued to Albanian citizens. The largest component is family reasons, followed by employment reasons, other reasons, and education reasons. This suggests that

migration is not only a labor-market issue; it also includes family reunification, student mobility, and other forms of long-term movement.

**Figure 3.** First residence permits issued to Albanian citizens in EU27 by reason, 2024. Source: Eurostat residence permits dataset.

**Table 4.** First residence permits issued to Albanian citizens in the EU27 by reason, 2024.

| Reason             | Permits in 2024 | Share of total EU27 permits |
|--------------------|-----------------|-----------------------------|
| Family reasons     | 34,017          | 51.4%                       |
| Employment reasons | 18,965          | 28.7%                       |
| Other reason       | 11,083          | 16.8%                       |
| Education reasons  | 2,096           | 3.2%                        |

## 7.5 Economic Indicators

Economic indicators show that Albania has improved in several areas, but migration pressure continues. GDP per capita increased from about \$1,160 in 2000 to about \$11,378 in 2024. Unemployment decreased from 19.0 percent in 2000 to 10.9 percent in 2025. However, net migration remains negative, which suggests that economic growth alone may not be enough to stop migration if wage levels, job quality, education opportunities, and career expectations remain stronger abroad.

**Table 5.** Selected economic indicators, 2000 to latest.

| Economic indicator            | 2000    | Latest value     | Basic interpretation                                 |
|-------------------------------|---------|------------------|------------------------------------------------------|
| GDP per capita current US\$   | \$1,160 | \$11,378 in 2024 | Income level increased strongly in nominal terms     |
| GDP growth annual %           | 7.5%    | 4.0% in 2024     | Growth continued but varies by year                  |
| Unemployment rate             | 19.0%   | 10.9% in 2025    | Labor-market indicator improved but remains relevant |
| Remittances received % of GDP | 16.7%   | 8.4% in 2024     | Remittances remain economically important            |
| FDI net inflows % of GDP      | 4.0%    | 6.3% in 2024     | Foreign investment is part of development context    |

## 8 Reasons Behind Brain Drain

The reasons behind Albania's brain drain are likely multi-dimensional. The study should not reduce migration to a single factor. Based on the data sources and migration literature context, the main reasons can be grouped as economic, educational, demographic, social, and institutional.

### 8.1 Economic Reasons

- Higher expected wages abroad compared with domestic income opportunities.
- Better job availability and career paths in destination countries.
- Informal work, unstable contracts, or lower job quality in the domestic labor market.
- Household strategies to increase income through migration and remittances.

### 8.2 Education and Human-Capital Reasons

- Students may leave for universities, training, or professional qualifications abroad.

- Young workers may see better skill-development opportunities outside Albania.
- Highly educated workers may migrate if local jobs do not match their education or career expectations.

### 8.3 Social and Network Reasons

- Family reunification can make migration easier and more stable.
- Existing diaspora communities reduce uncertainty and provide information, housing, and employment contacts.
- Migration can become a social norm in communities where many households already have migrants abroad.

## 9 Effects of Brain Drain on Albania

Brain drain can have both negative and positive effects. A balanced academic paper should show both sides, while still explaining why population decline and loss of human capital are serious concerns.

### 9.1 Negative Effects

- Reduction of the working-age population and possible labor shortages.
- Loss of educated young people and skilled workers.
- Pressure on pension and social-support systems as the elderly share increases.
- Regional imbalance if smaller towns and villages lose more young people than Tirana or larger urban areas.
- Lower domestic innovation capacity if professionals and students do not return.
- Family separation and social costs for children, parents, and elderly relatives left behind.

### 9.2 Positive or Potentially Positive Effects

- Remittances can support household consumption, education, housing, and poverty reduction.
- Return migrants can bring skills, experience, savings, and business ideas.
- Diaspora networks can support trade, investment, tourism, knowledge transfer, and professional links.
- Migration can reduce short-term unemployment pressure if local job creation is weak.

## 10 Discussion

The preliminary evidence supports the idea that Albania's brain drain is connected to long-term demographic decline and labor-market pressure. The population has declined significantly since 2000, net migration has remained negative, and residence permits show that migration is happening through multiple channels. Family migration is large, employment migration is important, and education migration is smaller in absolute numbers but highly relevant for human capital.

One important point is that migration is not automatically negative. Many Albanian households benefit from remittances, and migrants can gain skills abroad. The policy problem is how to reduce forced or unwanted migration, improve opportunities at home, and convert migration into development through return, investment, remote work, and diaspora cooperation.

The study should therefore avoid a simple conclusion such as 'migration is bad'. A better conclusion is that unmanaged brain drain can weaken Albania's long-term development, especially if young and working-age people leave faster than the country can create attractive jobs, competitive wages, and strong institutions.

## 11 Policy Recommendations

**Table 6.** Policy recommendations.

| Policy area                              | Recommendation                                                                                       |
|------------------------------------------|------------------------------------------------------------------------------------------------------|
| Improve job quality and wages            | Support sectors that can create skilled jobs, not only low-wage employment.                          |
| Strengthen youth employment programs     | Connect universities, vocational training, and employers through internships and practical skills.   |
| Create return-migration incentives       | Offer support for returnees who want to start businesses or use foreign experience in Albania.       |
| Use diaspora networks better             | Build platforms for diaspora investment, mentoring, remote collaboration, and professional exchange. |
| Reduce information gaps before migration | Provide reliable pre-departure information about costs, wages, contracts, and rights abroad.         |
| Improve regional development             | Create opportunities outside Tirana to reduce regional population loss.                              |
| Monitor brain drain with data            | Build regular dashboards using migration, labor-market, education, and demographic indicators.       |

## 12 Limitations

- This is a low-level academic draft and does not yet include full statistical testing.
- Eurostat immigration into European reporting countries is a proxy, not a direct measure of emigration from Albania.
- Some destination-country data may be missing or incomplete in the extracted Eurostat files.
- Residence permit categories must be handled carefully to avoid double counting TOTAL values with subcategories.
- Correlation analysis cannot prove causation.
- Forecasts to 2035 depend on assumptions and should be interpreted as scenarios, not guaranteed outcomes.
- GDP and unemployment may not fully capture wage quality, corruption perceptions, education quality, or personal migration motives.

## 13 Conclusion

This paper provides a structured low-level research draft on Albania's brain drain. The preliminary evidence shows that the issue is real and important: Albania has experienced a significant population decline since 2000, net migration remains negative, the working-age and youth population base is shrinking, and residence permits show active migration channels linked to family, employment, education, and other reasons.

The main conclusion is that Albania's brain drain should be studied as both a demographic and economic challenge. It affects labor supply, population ageing, household income, public policy, and long-term development. The next stage of the project should clean the datasets, build the database, perform exploratory and statistical analysis, create forecasts to 2035, and present results in a Power BI dashboard and final research report.

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